

Ground-state uniqueness on convex domains and sharp logarithmic concavity on balls

Henry Zweiman

September 16, 2026

Abstract

We prove uniqueness of the positive Dirichlet ground state of $-\Delta u = u \log u^2$ on every bounded convex domain, without boundary smoothness assumptions. Every positive log-concave solution satisfies $D^2(-\log u) \succ I$ and is this ground state. A positive-level regularization gives the Hessian bound, while a local strict transport remainder in the logarithmic Sobolev inequality yields a sharp energy comparison on H_0^1 with equality precisely for multiples of the ground state. This answers the ground-state uniqueness and identification questions of Gallo, Mosconi and Squassina, whose log-concave existence theorem is an essential input. On balls, we prove a globally convergent positive radial series for $-\log(u/u(0))$, with an explicit recurrence. It yields concavity of $-\sqrt{-\log(u/\|u\|_\infty)}$, the sharp exponent $1/2$ among powers of the negative normalized logarithm, and positive power concavity of u . We also prove that the normalized square-root-log property fails under arbitrarily small smooth uniformly convex deformations near every ball in dimensions at least two. A degree-three shape derivative creates a cubic-jet obstruction at the moving maximum. The deformed solutions remain strictly log-concave and are therefore ground states. Uniqueness among all positive solutions remains open here.

Mathematics Subject Classification (2020). 35B45, 35J61, 34B15, 26A51.

Keywords. Ground-state uniqueness; convex domains; logarithmic Sobolev inequality; logarithmic Schrödinger equation; concavity; radial solutions; positive power series; shooting method.

1. Introduction and main results

Let $N \geq 1$, $R > 0$, and $B_R = \{x \in \mathbb{R}^N : |x| < R\}$. We consider

$$-\Delta u = u \log u^2 \quad \text{in } B_R, \quad u > 0 \quad \text{in } B_R, \quad u = 0 \quad \text{on } \partial B_R, \quad (1.1)$$

where $u \in C^2(B_R) \cap C(\overline{B_R})$. All concavity statements about a logarithmic transform concern the open ball. Statements about u^α extend to the closed ball by continuity.

Gallo, Mosconi and Squassina [GMS] establish existence of a solution with locally negative-definite logarithmic Hessian on every bounded convex domain. Their Theorem 6.5 gives radial symmetry and radial decrease of every positive solution on a ball. In Sections 6.2–6.3 they ask for stronger concavity in balls: positive power concavity of u , and concavity of the negative square root of its negative normalized logarithm. Their Theorem 6.6 proves both properties in dimension one. The second question remains explicitly stated on page 44 of the final article.

The positive-series representation below answers both ball questions in every dimension. Theorem 1.5 additionally resolves the ground-state questions of [GMS, Remark 1.6] on every bounded convex domain. Write $u(x) = U(r)$, $r = |x|$, and set

$$M = \log U(0), \quad w(r) = -\log \frac{U(r)}{U(0)}, \quad A = \frac{2M}{N}. \quad (1.2)$$

Theorem 1.1 (Positive radial series). Every solution of (1.1) satisfies $A > 1$. There are strictly positive coefficients c_n , $n \geq 0$, such that

$$\frac{w'(r)}{r} = \sum_{n=0}^{\infty} c_n r^{2n}, \quad w(r) = \sum_{j=1}^{\infty} \frac{c_{j-1}}{2j} r^{2j}, \quad 0 \leq r < R. \quad (1.3)$$

The quotient at $r = 0$ is interpreted by continuity. Both series, regarded as series in r^2 , have radius of convergence exactly R^2 . Their coefficients are determined by

$$c_0 = A, \quad c_{n+1} = \frac{(n+1) \sum_{i=0}^n c_i c_{n-i} - c_n}{(n+1)(2n+N+2)}. \quad (1.4)$$

In particular, $Z(t) = w'(\sqrt{t})/\sqrt{t}$ is absolutely monotone on $[0, R^2]$: every derivative is strictly positive there. The same is true of every derivative of positive order of $W(t) = w(\sqrt{t})$. In addition,

$$D^2 w(|x|) \succeq AI_N, \quad (1.5)$$

with strict inequality as quadratic forms when $x \neq 0$. Here $\succeq$ denotes the order on symmetric matrices.

Theorem 1.2 (Sharp transformed concavity). For every solution of (1.1),

$$x \mapsto -\left(-\log \frac{u(x)}{\|u\|_{\infty}}\right)^{\beta} \text{ is concave on } B_R \iff \beta \geq \frac{1}{2}, \quad \beta > 0. \quad (1.6)$$

For $\beta = 1/2$ its Hessian is negative definite away from the center. At the center it is continuous and has a norm-type cusp; no differentiability assertion there is intended.

Theorem 1.3 (Positive power concavity). There exists a number $\alpha_*(U) \in (0, 1)$ such that, for $\alpha > 0$,

$$u^{\alpha} \text{ is concave on } \overline{B_R} \iff \alpha \leq \alpha_*(U). \quad (1.7)$$

It is characterized by an attained interior minimum,

$$\alpha_*(U) = \min_{0 < r < R} \frac{w''(r)}{w'(r)^2}. \quad (1.8)$$

The formula is implicit. We do not assert a positive exponent uniform over all radii, or monotonicity of this optimal exponent as a function of the radius.

Theorem 1.4 (Instability under convex deformation). Let $N \geq 2$, $R > 0$, and let U be the positive Dirichlet solution on B_R . Define $Y_3(\theta) = \theta_1^3 - 3\theta_1\theta_2^2$ and

$$\Omega_\varepsilon = \{r\theta : 0 \leq r < R + \varepsilon Y_3(\theta), \theta \in S^{N-1}\}.$$

For all sufficiently small $|\varepsilon|$, these are smooth uniformly convex domains and there is a smooth branch of positive solutions u_ε , with $u_0 = U$. For each sufficiently small nonzero ε , the solution is strictly log-concave and has a unique maximum, but

$$-\sqrt{-\log(u_\varepsilon/\|u_\varepsilon\|_\infty)}$$

is not concave in any neighborhood of that maximum. The domains converge to B_R in every C^k boundary norm.

The normalization in this statement is essential. The scale-invariant notion of $1/2$ -log-concavity permits a sufficiently small multiplicative factor in front of u ; see [IST, Section 4.2]. Theorem 1.4 concerns the specific factor $1/\|u\|_\infty$, where the transform has its cusp at the maximum. It does not disprove the weaker rescaled property. It answers the controlled-deformation question posed in the earlier version of Section 7 and leaves the ball theorem unchanged.

Theorem 1.5 (Ground-state uniqueness on convex domains). Let $\Omega \subset \mathbb{R}^N$ be nonempty, bounded, open and convex. The problem $-\Delta U = U \log U^2$ with zero Dirichlet boundary values has a unique positive ground state U . Every positive log-concave solution is this ground state, and

$$D^2(-\log U)(x) \succ I_N \quad (x \in \Omega).$$

Let $Z = \int_\Omega U^2$. For every nonzero real $v \in H_0^1(\Omega)$,

$$\int_\Omega |\nabla v|^2 - \int_\Omega v^2 \log v^2 + \|v\|_2^2 \log \frac{\|v\|_2^2}{Z} \geq 0,$$

with equality exactly for the nonzero multiples of U . The real ground states are $\pm U$. No regularity of $\partial\Omega$ is required. Here a ground state minimizes the action $\mathcal{J}(v) = \frac{1}{2} \int (|\nabla v|^2 + v^2 - v^2 \log v^2)$ on its nonzero Nehari set; equivalently it is a nonzero solution of least action.

This theorem does not assert uniqueness among all positive solutions. It identifies the solution constructed in [GMS] as the ground state and proves its uniqueness at least energy. Its proof in Section 8 uses interior level sets and a local strict entropy remainder to avoid assumptions on boundary derivatives or global integrability of quotients of solutions.

1.1. Prior results and the scope of the contribution

The amplitude threshold, uniqueness in balls, and strict ordering of first zeros are prior results. For $k = 1$ and $\beta = 0$, Ben Chrouda [BC, Theorem 1.2 and Corollaries 1.3 and 1.5] proves these facts for $-\Delta v = v \log v$. The change $v(y) = u(y/\sqrt{2})$ gives the normalization (1.1). The equality case in the amplitude threshold is excluded by the Gaussian solution and uniqueness of the central initial-value problem. We give new proofs of these particular consequences because they fit the recurrence argument, but do not claim their first discovery.

Liu, Sun and Zou [LSZ] prove a broader shooting classification of whole-space nodal solutions; in particular, their argument establishes strict motion of zeros with the initial value. Gallo, Moraschi and Squassina [GMoS] develop parabolic concavity principles whose logarithmic example yields preservation

of logarithmic concavity. Neither of these results supplies the positive radial series used here. For the ball results, our priority claim concerns that structure and the resulting stronger concavity, rather than existence, radial symmetry, whole-space Gaussian ground-state classification, or shooting uniqueness. Current versions of these sources were compared on September 15, 2026. The convex-domain ground-state extension was checked against the final [GMS] article and later whole-space uniqueness results on September 16, 2026. Its contribution is the Hessian strengthening and the resulting variational rigidity on arbitrary bounded convex domains. Caffarelli contraction, Gaussian logarithmic Sobolev inequalities and interior elliptic regularity are classical inputs, credited in Section 8.

The proof of Theorem 1.1 has three stages. An integrating factor puts the logarithmic radial derivative above its Gaussian value. Its equation then produces positive Taylor coefficients. Finally, positivity and a real-axis singularity argument show that the series represents the solution all the way to its first zero. Convexity of a Hilbert-space norm proves Theorem 1.2. A separate boundary argument, which accommodates the singular derivative of $u \log u^2$ at zero, completes Theorem 1.3. For Theorem 1.4, the ball Hessian bound and the classical weighted Poincare inequality make the linearized Dirichlet operator invertible. A degree-three shape derivative then violates a necessary cubic-jet condition at the moving maximum.

2. The radial equation and the Gaussian threshold

We use radial symmetry and radial decrease from [GMS, Theorem 6.5]. These are nontrivial inputs: the derivative of the nonlinearity is unbounded below near zero, so an unqualified appeal to the usual locally Lipschitz moving-plane theorem would be inappropriate. Existence follows from [GMS, Theorem 1.1], with its one-dimensional result covering $N = 1$. The rest of the argument applies to every positive radial decreasing solution.

Put $f(s) = s \log s^2$ for $s > 0$, and $f(0) = 0$. The radial equation and its integral form are

$$U'' + \frac{N-1}{r}U' + f(U) = 0, \quad U'(r) = -r^{1-N} \int_0^r s^{N-1} f(U(s)) ds. \quad (2.1)$$

The central initial-value problem is unique as long as its solution stays positive. Indeed, integrating (2.1) once more gives a Volterra equation. On a sufficiently small interval the Lipschitz bound for f near a prescribed positive central value makes its integral operator a contraction, since the norm of the double integral is bounded by a constant times r^2 . Ordinary uniqueness then continues the solution away from zero. The same identity, using smoothness of f near a positive central value, gives the usual even expansions at the center.

Since U has its maximum at zero, (2.1) implies $f(U(0)) \geq 0$, hence $M \geq 0$. If $M = 0$, central uniqueness would give $U \equiv 1$, which violates the boundary condition. Thus $M > 0$.

In fact $U'(r) < 0$ for $0 < r < R$. If U' vanished there, its nonpositivity would imply $U'' = 0$ at that point. Equation (2.1) would give $U = 1$, and ordinary uniqueness with data $(U, U') = (1, 0)$ would again force the constant solution.

In terms of (1.2), equation (2.1) becomes

$$w'' + \frac{N-1}{r}w' - (w')^2 = 2M - 2w. \quad (2.2)$$

Define $z(r) = w'(r)/r$ for $r > 0$ and extend it to the center. The central expansion gives

$$z(0) = \frac{2M}{N} = A, \quad z'(r) = O(r). \quad (2.3)$$

Differentiation of (2.2) yields

$$z'' + \left(\frac{N+1}{r} - 2rz \right) z' = 2z(z-1). \quad (2.4)$$

Equivalently,

$$\left(r^{N+1} e^{-2w} z' \right)' = 2r^{N+1} e^{-2w} z(z-1). \quad (2.5)$$

Lemma 2.1. We have $A > 1$ and $z'(r) > 0$ for all $0 < r < R$.

Proof. Suppose first that $0 < A < 1$. Integrating (2.5) from zero shows that $z' < 0$ while $0 < z < 1$. The boundary term at zero vanishes by (2.3). The function cannot first reach 1, since it is decreasing. It cannot first reach zero either: the integral in (2.5) would then give a strictly negative derivative, whereas $z = w'/r > 0$ throughout the interior. Thus $0 < z \leq A$ on $[0, R)$, giving $w(r) \leq Ar^2/2$. This contradicts $w(r) \rightarrow \infty$ as $r \uparrow R$.

If $A = 1$, the central solution is

$$U(r) = \exp \left(\frac{N}{2} - \frac{r^2}{2} \right), \quad (2.6)$$

by the uniqueness following (2.1). It has no finite zero. Hence $A > 1$. Now (2.5) gives $z' > 0$ as long as $z > 1$, and this strict increase prevents any first exit from that region. $\square$

The eigenvalues of $D^2 w(|x|)$ are $w'' = z + rz'$ in the radial direction and z in the tangential directions. Lemma 2.1 proves (1.5), including its strict form away from the center. In dimension one only the radial eigenvalue is present.

3. A positive recurrence valid throughout the ball

Let $t = r^2$ and $Z(t) = z(\sqrt{t})$. Equation (2.4) becomes

$$2tZ'' + (N+2)Z' - 2tZZ' = Z(Z-1). \quad (3.1)$$

Lemma 3.1 (Local construction). For each $A > 1$, recurrence (1.4) defines strictly positive coefficients satisfying

$$0 < c_n \leq A^{n+1}, \quad n \geq 0. \quad (3.2)$$

The resulting series solves (3.1) near zero and agrees with the transformed radial solution.

Proof. The coefficient of t^n in the linear derivative terms of (3.1) is $(n+1)(2n+N+2)c_{n+1}$. In the nonlinear terms, symmetry of convolution gives

$$[t^n](Z^2 + 2tZZ') = (n+1) \sum_{i=0}^n c_i c_{n-i}, \quad (3.3)$$

which proves (1.4). In particular $c_1 = A(A-1)/(N+2) > 0$. For $n \geq 1$, its numerator can be written

$$(2(n+1)A-1)c_n + (n+1) \sum_{i=1}^{n-1} c_i c_{n-i}, \quad (3.4)$$

so positivity follows by induction. Dropping the negative term in (1.4) and using the inductive upper bounds gives

$$c_{n+1} \leq \frac{(n+1)A^{n+2}}{2n+N+2} \leq A^{n+2}. \quad (3.5)$$

Consequently the series converges for $|t| < 1/A$, and termwise differentiation verifies (3.1).

Integrate $w'(r) = rZ(r^2)$ with $w(0) = 0$. The derivative of the difference between the two sides of (2.2) vanishes by (3.1). Its limiting value at zero is $NA - 2M = 0$. Thus (2.2) holds. Setting $U(r) = e^{M-w(r)}$ recovers the radial equation with the prescribed central data. The uniqueness argument after (2.1) identifies this constructed solution with the given one. $\square$

We next state the positive-series continuation principle, including its short proof.

Lemma 3.2 (Positive real singularity). Suppose $F(t) = \sum_{n \geq 0} a_n t^n$ has $a_n \geq 0$ and finite positive radius of convergence ρ . It cannot admit a holomorphic continuation to a neighborhood of the positive real point ρ .

Proof. If such a continuation existed, continuity of its derivatives and monotone convergence as $t \uparrow \rho$ would give, for every $k \geq 0$,

$$F^{(k)}(\rho) = \sum_{n \geq k} \frac{n!}{(n-k)!} a_n \rho^{n-k}. \quad (3.6)$$

Choose a sufficiently small $h > 0$ in the convergence disk of the Taylor series centered at ρ . Its coefficients are nonnegative by (3.6). Interchanging nonnegative sums gives

$$\begin{aligned} \sum_{k \geq 0} \frac{F^{(k)}(\rho)}{k!} h^k &= \sum_{n \geq 0} a_n \sum_{k=0}^n \binom{n}{k} \rho^{n-k} h^k \\ &= \sum_{n \geq 0} a_n (\rho + h)^n < \infty, \end{aligned} \quad (3.7)$$

contradicting the definition of ρ . $\square$

Proof of Theorem 1.1. By Lemma 3.1 the series for Z represents the actual solution near zero. Denote its convergence radius by ρ . If $\rho < R^2$, the actual solution is regular at the positive point ρ . On an interval around that point, (3.1) is an ordinary differential equation with analytic right-hand side after solving for Z'' . It therefore has a locally holomorphic solution with the actual values of Z and Z' there. Ordinary uniqueness identifies this solution with the continuation from zero. This contradicts Lemma 3.2. Thus $\rho \geq R^2$.

Conversely, $z(r)$ tends to infinity as $r \uparrow R$: by Lemma 2.1 it is increasing, and a finite limit would make $w(r) = \int_0^r sz(s) ds$ bounded. Therefore $\rho > R^2$ is impossible. We have proved $\rho = R^2$. Integration gives the second series in (1.3). Dividing its coefficients by a factor linear in the index leaves the convergence radius unchanged. All derivative assertions now follow by differentiating convergent positive series on compact subintervals of $[0, R^2]$. The Hessian assertion was proved after Lemma 2.1. $\square$

The continuation step is essential: positivity of finitely many coefficients, or even a formal positive recurrence without convergence control, would not imply a global concavity theorem.

4. Convexity of the square root and sharpness

Put $a_j = c_{j-1}/(2j) > 0$. Theorem 1.1 gives

$$\sqrt{w(r)} = \left\| (\sqrt{a_j} r^j)_{j \geq 1} \right\|_{\ell^2}, \quad 0 \leq r < R. \quad (4.1)$$

Every coordinate on the right is nonnegative, nondecreasing and convex as a function of r . Coordinatewise convexity, monotonicity of the ℓ^2 norm on nonnegative sequences, and the triangle inequality show that $r \mapsto \sqrt{w(r)}$ is convex and nondecreasing. Equivalently one may first prove this for the finite partial sums and pass to the finite pointwise limit on $[0, R)$.

For strictness it is useful to give an exact identity. At $r > 0$ let

$$p_j = \frac{a_j r^{2j}}{w(r)}, \quad m = \sum_{j \geq 1} j p_j, \quad v = \sum_{j \geq 1} (j - m)^2 p_j. \quad (4.2)$$

These are a probability distribution, its mean and its variance; all moments are finite inside the convergence interval. Direct differentiation gives

$$\frac{r^2(2ww'' - (w')^2)}{4w^2} = 2v + m(m - 1) > 0. \quad (4.3)$$

The strict inequality holds because $m > 1$, as $a_2 > 0$. Thus $(\sqrt{w})'' > 0$ for $r > 0$. Its tangential Hessian eigenvalue is $(\sqrt{w})'/r > 0$.

Proof of Theorem 1.2. A convex nondecreasing function of the Euclidean norm is convex. Applying this fact to (4.1) proves convexity of $x \mapsto \sqrt{w(|x|)}$ across the center as well. The Hessian assertions away from the center follow from (4.3). For $\beta \geq 1/2$, the function $s \mapsto s^{2\beta}$ is convex and nondecreasing on $[0, \infty)$; composing it with $\sqrt{w}$ proves the forward implication of (1.6).

For necessity, the central expansion is

$$w(r) = \frac{A}{2} r^2 + O(r^4). \quad (4.4)$$

If $0 < \beta < 1/2$, the restriction $w(r)^\beta$ to a positive radial segment has strictly negative second derivative for sufficiently small $r > 0$. Indeed its leading term is $(A/2)^\beta r^{2\beta}$, whose second derivative has the negative factor $2\beta(2\beta - 1)$; the differentiated remainder has higher order. It cannot be convex. Finally, (4.4) gives $\sqrt{w(|x|)} = \sqrt{A/2}|x| + O(|x|^3)$, which describes the stated cusp. $\square$

The same series gives more than a Hessian bound. For every integer $k \geq 1$, the radial differential operator $\mathcal{D} = (2r)^{-1}d/dr$ satisfies $\mathcal{D}^k w(r) > 0$ on $0 < r < R$, with a positive limit at zero. These infinitely many sign inequalities refer to differentiation with respect to squared radius, not arbitrary mixed spatial derivatives.

5. Boundary behavior and the optimal positive power

Because f is bounded on the compact range of U , (2.1) gives a finite limit of U' at R . The differential equation then gives a continuous limit of U'' there. The next argument rules out a zero boundary slope without assuming a finite value of $f'(0)$.

Lemma 5.1 (Simple boundary zero). There is $b > 0$ such that

$$U(r) = b(R - r) + O((R - r)^2), \quad U'(r) = -b + O(R - r), \quad (5.1)$$

and

$$z(r) \sim \frac{1}{R(R - r)} \quad \text{as } r \uparrow R. \quad (5.2)$$

Proof. Write $v = -U' > 0$. Near R we have $0 < U < 1$, and (2.1) gives $U'' > 0$. Thus v is decreasing there. Set

$$F(s) = \frac{s^2}{2}(\log s^2 - 1), \quad E(r) = \frac{v(r)^2}{2} + F(U(r)). \quad (5.3)$$

The energy identity is $E' = -(N - 1)v^2/r$. Suppose $U'(R) = 0$. Then $E(R) = 0$, and for r sufficiently close to R ,

$$E(r) = \int_r^R \frac{N - 1}{s} v(s)^2 ds \leq C(R - r)v(r)^2. \quad (5.4)$$

Choose this neighborhood so that $C(R - r) \leq 1/4$. Equation (5.4) implies

$$\frac{v(r)^2}{4} \leq -F(U(r)), \quad v(r) \leq C_1 U(r) \sqrt{\log(1/U(r))}. \quad (5.5)$$

Since U is strictly decreasing, it can be used as the variable of integration on this interval. The remaining time to reach zero would satisfy

$$R - r = \int_0^{U(r)} \frac{ds}{v(U^{-1}(s))} \geq \frac{1}{C_1} \int_0^{U(r)} \frac{ds}{s \sqrt{\log(1/s)}} = \infty, \quad (5.6)$$

a contradiction. When $N = 1$, the integral in (5.4) is zero and the same argument applies. Thus $b = -U'(R) > 0$. The bounded second derivative gives (5.1), and $z = -U'/(rU)$ gives (5.2). $\square$

Proof of Theorem 1.3. For $r > 0$,

$$(U^\alpha)'' = \alpha U^\alpha (\alpha (w')^2 - w''), \quad \frac{(U^\alpha)'}{r} = -\alpha U^\alpha \frac{w'}{r} < 0. \quad (5.7)$$

Thus the radial Hessian condition is precisely $\alpha \leq q(r)$, where $q = w''/(w')^2$. The condition at the center holds automatically, because $D^2 u^\alpha(0) = -\alpha U(0)^\alpha A I_N$. By Lemma 2.1, $q > 0$. Equation (4.4) gives $q(r) \rightarrow \infty$ at zero, while Lemma 5.1 gives

$$q(r) = 1 - \frac{U(r)U''(r)}{U'(r)^2} \longrightarrow 1 \quad \text{as } r \uparrow R. \quad (5.8)$$

Moreover $U'' > 0$ near R , so $q < 1$ there. Choose any such point. The limits at both endpoints and continuity imply that the global infimum is achieved on a compact interior interval; its value is strictly between zero and one. Equations (5.7) prove (1.7)–(1.8), and continuity extends concavity to the closed ball. $\square$

6. Coefficient comparison and two consequences

The recurrence contains information about variation with the central height. Fix N and put $\delta = A - 1 > 0$. Write $c_n(\delta)$ for its coefficients.

Proposition 6.1 (Positive parameter dependence). For every $n \geq 1$, $c_n(\delta)$ is a polynomial in δ with nonnegative coefficients and zero constant term. If two solutions have $0 < \delta_1 < \delta_2$ and $\theta = \delta_2/\delta_1$, then on their common radial interval,

$$z_2(r) - 1 \geq \theta(z_1(r) - 1), \quad (6.1)$$

and

$$w_2(r) - \frac{r^2}{2} \geq \theta \left(w_1(r) - \frac{r^2}{2} \right). \quad (6.2)$$

Proof. We have $c_1 = (\delta + \delta^2)/(N + 2)$. For $n \geq 1$, rewrite (1.4) as

$$c_{n+1} = \frac{(2n + 1 + 2(n + 1)\delta)c_n + (n + 1) \sum_{i=1}^{n-1} c_i c_{n-i}}{(n + 1)(2n + N + 2)}. \quad (6.3)$$

The polynomial assertion follows by induction. Every monomial of positive degree satisfies $\delta_2^k \geq \theta \delta_1^k$. Apply this coefficientwise, including the constant-in- r term $c_0 - 1 = \delta$, and sum the convergent series to obtain (6.1). Integrating $r(z_i(r) - 1)$ gives (6.2). $\square$

Corollary 6.2 (Known shooting order and uniqueness). If $U_2(0) > U_1(0)$, their first zero radii satisfy $R_2 < R_1$. Consequently (1.1) has exactly one positive solution for every $R > 0$.

Proof. Suppose $R_2 \geq R_1$. By (6.1) and divergence of z_1 at R_1 , strict inequality $R_2 > R_1$ is impossible. If $R_2 = R_1$, (5.2) gives $z_2/z_1 \rightarrow 1$, contradicting (6.1) and $\theta > 1$. This proves the strict shooting order. Two distinct positive solutions on the same ball are radial and have either different central values, contradicted by this order, or equal central values, contradicted by initial-value uniqueness. Existence is the prior input stated in Section 2. $\square$

As emphasized in Section 1.1, this is an alternative proof of a prior theorem. The coefficientwise parameter comparison (6.1)–(6.2) is the additional conclusion of the present method.

Corollary 6.3 (Products of balls). Let u_i be positive solutions of (1.1) on $B_{R_i} \subset \mathbb{R}^{N_i}$, for $1 \leq i \leq k$. Then

$$u(x_1, \dots, x_k) = \prod_{i=1}^k u_i(x_i) \quad (6.4)$$

solves the same equation on the product domain, and $-\sqrt{-\log(u/\|u\|_\infty)}$ is concave there.

Proof. Dividing the Laplacian of the product by the product gives $-\Delta u/u = \sum_i \log u_i^2 = \log u^2$. The negative normalized logarithm is $\sum_i w_i$. Its square root is the Euclidean norm of the vector $(\sqrt{w_i})_{i=1}^k$. Each component is nonnegative and convex in its corresponding variables by Theorem 1.2. The argument of (4.1) proves convexity of this norm. $\square$

This includes the interval-product construction already treated in [GMS]; the factorization itself is elementary and is not a separate novelty claim.

7. Instability under convex deformation

We prove Theorem 1.4. All boundary-distance and function-space statements below refer to a fixed smooth ball before pulling back the domain.

7.1. A necessary cubic-jet condition

Let $w \in C^4$ near x_0 , with $w(x_0) = 0$, $Dw(x_0) = 0$ and positive-definite Hessian $H = D^2w(x_0)$. If $\sqrt{w}$ is convex in a neighborhood of x_0 , then

$$D^3w(x_0) = 0. \quad (\text{D1})$$

Indeed, on any unit direction e , write

$$w(x_0 + te) = at^2 + bt^3 + O(t^4), \quad a = \frac{1}{2}H[e, e] > 0, \quad b = \frac{1}{6}D^3w(x_0)[e, e, e].$$

For $t > 0$, the second derivative of the square root tends to $b/\sqrt{a}$; for $t < 0$ it tends to $-b/\sqrt{a}$. To justify differentiating the remainder, write $w(x_0 + te) = t^2h(t)$ with h twice continuously differentiable and $h(0) = a$, $h'(0) = b$. Convexity on both half intervals forces $b = 0$. A symmetric cubic tensor whose diagonal polynomial vanishes is zero. Conversely, if the tensor is nonzero, one half of a line has strictly negative square-root second derivative at points arbitrarily close to x_0 . The vanishing condition is necessary, not sufficient.

7.2. The ball linearization is invertible

Use Theorem 1.1 and its estimate

$$-D^2 \log U \succeq AI_N, \quad A = \frac{2 \log U(0)}{N} > 1. \quad (\text{D2})$$

The linearized Dirichlet operator is

$$L = -\Delta - (2 \log U + 2). \quad (\text{D3})$$

It satisfies $LU = -2U$. Since $U > 0$, -2 is its simple first eigenvalue. For every $\phi \in H_0^1(B_R)$ perpendicular to U in L^2 ,

$$\langle L\phi, \phi \rangle \geq (2A - 2)\|\phi\|_2^2. \quad (\text{D4})$$

Here U has a strictly negative outward derivative and U is comparable to the boundary distance d . Consequently the logarithmic potential is form-bounded with relative bound zero on H_0^1 : use

$|\log d| \leq \eta d^{-2} + C_\eta$ and Hardy's inequality. The operator has compact resolvent and its form domain is H_0^1 .

For a compactly supported smooth ϕ , set $h = \phi/U$. Integration by parts gives the ground-state identity

$$\langle L\phi, \phi \rangle = \int U^2 |Dh|^2 - 2 \int U^2 h^2. \quad (\text{D5})$$

The classical strongly log-concave weighted Poincare inequality (see [KS, Theorem 2.1] for the Brascamp-Lieb form on a convex domain), applied to the density U^2 , gives

$$\int U^2 |h - \bar{h}|^2 \leq \frac{1}{2A} \int U^2 |Dh|^2, \quad \bar{h} = \frac{\int U^2 h}{\int U^2}. \quad (\text{D6})$$

The relevant boundary version can be obtained without any assumption on a positive limiting density at ∂B_R . On each smaller concentric ball, the density is smooth and positive. For its weighted Neumann operator $\mathcal{L} = \Delta + 2D \log U \cdot D$, integration of the Bochner identity for an eigenfunction gives

$$\int (\mathcal{L}h)^2 U^2 = \int (|D^2 h|^2 - 2D^2 \log U [Dh, Dh]) U^2 + \int_{\partial B_r} \text{II}[D_T h, D_T h] U^2. \quad (\text{D7})$$

The outward second fundamental form II of the sphere is positive. Equation (D2) therefore bounds each nonconstant weighted Neumann eigenvalue below by $2A$. The spectral expansion yields (D6) on B_r ; let $r \uparrow R$. First apply this to the smooth compactly supported quotient. Ground-state identity and form density then extend the result to all H_0^1 functions. Perpendicularity to U is exactly $\bar{h} = 0$, proving (D4). Thus L has no zero eigenvalue and has exactly one negative eigenvalue.

For use in a shape argument, fix $0 < \gamma < 1$ and set $X = \{v \in C^{2,\gamma}(\bar{B}_R) : v|_{\partial B_R} = 0\}$, $Y = C^{0,\gamma}(\bar{B}_R)$. Then

$$L : X \longrightarrow Y \text{ is an isomorphism.} \quad (\text{D8})$$

To see the functional-analytic point at the vanishing boundary, choose a smooth positive defining function ρ equal to d near the boundary. Division by ρ maps X continuously to $C^{1,\gamma}$, and $U = \rho a$ with $a > 0$ on the closed ball. Hence multiplication by $\log U$ maps X to Y . It is compact: for $\gamma < \beta < 1$, a bounded sequence in X has a subsequence converging in $C^{1,\beta}$; its quotients by ρ converge in $C^{0,\beta}$, and multiplication by $\rho \log \rho$ maps that convergence to $C^{0,\gamma}$. Thus L is a compact perturbation of the Dirichlet Laplacian isomorphism. Its kernel is zero by (D4) and $LU = -2U$; the Fredholm alternative proves (D8).

7.3. A genuine branch on deformed domains

Let

$$H_3(x) = x_1^3 - 3x_1x_2^2, \quad Y_3 = H_3|_{S^{N-1}},$$

and define

$$\Omega_\varepsilon = \{r\theta : 0 \leq r < R + \varepsilon Y_3(\theta), \theta \in S^{N-1}\}. \quad (\text{D9})$$

For all sufficiently small $|\varepsilon|$, this is a smooth uniformly convex domain. Indeed the radial graph converges to the sphere in C^2 and its principal curvatures converge uniformly to $1/R > 0$; its outward normal parametrization, or the usual support-plane criterion, gives convexity. It converges in every C^k norm because Y_3 is smooth.

Choose a smooth radial cutoff χ zero on $B_{R/2}$ and one near the boundary, and use

$$T_\varepsilon(x) = x + \varepsilon \chi(|x|) Y_3(x/|x|) \frac{x}{|x|}, \quad (\text{D10})$$

with the second term zero near the origin. For small parameter this is a smooth diffeomorphism from the ball to (D9), equal to the identity near zero. The pulled-back Laplacian has coefficients smooth in x and real analytic in ε .

Although $s \log s^2$ is not differentiable at $s = 0$ as a scalar function, the Dirichlet nonlinear map is smooth in the spaces just specified near U . On the open positive-Hopf cone write $v = \rho b$, $b > 0$; then

$$v \log v^2 = 2\rho b \log \rho + 2\rho b \log b. \quad (\text{D11})$$

The first term is a bounded linear map of b into Y , and the second is real analytic for b bounded away from zero in $C^{1,\gamma}$. Division by ρ is bounded on X . Thus this is a real-analytic map from that open cone in X to Y . At parameter zero, its equation has derivative (D8). The Banach implicit-function theorem produces a unique nearby branch $v_\varepsilon \in X$ with $v_0 = U$ satisfying the pulled-back equation. It remains in the positive-Hopf cone. Set $u_\varepsilon = v_\varepsilon \circ T_\varepsilon^{-1}$ on Ω_ε ; these are genuine positive Dirichlet solutions. Interior elliptic regularity gives smooth dependence in every local C^k norm. This does not assert global uniqueness among all positive solutions on the deformed domains.

7.4. The shape derivative has a nonzero cubic term

Let $V = \partial_\varepsilon T_\varepsilon|_0$ and let z be the Eulerian shape derivative, $z = \dot{v}_0 - DU \cdot V$. Differentiation of the equation in each fixed compact subset and of the transported zero boundary value gives

$$Lz = 0 \text{ in } B_R, \quad z|_{\partial B_R} = -U'(R)Y_3. \quad (\text{D12})$$

This is a weak Dirichlet problem with a smooth nonzero boundary trace; z need not be C^2 up to the boundary, since the potential is logarithmic there. It is $C^{1,\gamma}$ up to the boundary and smooth in the interior. Uniqueness follows from (D8), or from the form kernel exclusion. Rotation invariance and spherical harmonic projection therefore give

$$z(r\theta) = \psi(r)Y_3(\theta),$$

where

$$-\psi'' - \frac{N-1}{r}\psi' + \frac{3(N+1)}{r^2}\psi - (2\log U + 2)\psi = 0, \quad \psi(R) = -U'(R) > 0. \quad (\text{D13})$$

Near zero, the regular solutions of this equation form a one-dimensional space and have the expansion

$$\psi(r) = cr^3 + O(r^5). \quad (\text{D14})$$

This follows by inserting $\psi = r^3 a(r)$: a satisfies a regular radial integral equation in dimension $N+6$, with smooth even potential near zero. Its initial value $a(0) = c$ determines the solution uniquely. In particular $c \neq 0$, since $c = 0$ would force $\psi = 0$, contradicting its boundary value. In fact $c > 0$: test the radial equation with its negative part times the same degree-three harmonic; (D4) is positive on this angular subspace, so that negative part vanishes, and uniqueness excludes a zero leading value. Thus

$$z(x) = cH_3(x) + O(|x|^5), \quad c > 0. \quad (\text{D15})$$

7.5. Move the maximum before computing its jet

On a small fixed ball at zero, D^2U is negative definite. The nearby solutions have the same property there. Outside a smaller neighborhood of zero, U has a strict value gap below $U(0)$, which persists under the uniform pullback convergence. Hence each nearby u_ε has a unique global maximum x_ε near zero.

The implicit-function theorem applied to the gradient gives smooth x_ε . Since $Dz(0) = 0$, differentiation gives $x'_0 = 0$, and therefore $x_\varepsilon = O(\varepsilon^2)$. Because the base solution is radial, $D^3U(0) = 0$. Interior differentiability and (D15) imply

$$D^3u_\varepsilon(x_\varepsilon) = \varepsilon c D^3H_3 + O(\varepsilon^2). \quad (\text{D16})$$

Set

$$w_\varepsilon(x) = -\log \frac{u_\varepsilon(x)}{u_\varepsilon(x_\varepsilon)}.$$

At the maximum the logarithmic third derivative has no lower-derivative cross terms, since $Du_\varepsilon = 0$. In particular,

$$D^3w_\varepsilon(x_\varepsilon)[e_1, e_1, e_1] = -\frac{6c}{U(0)}\varepsilon + O(\varepsilon^2) \neq 0 \quad (\text{D17})$$

for every sufficiently small nonzero parameter. Its Hessian at the maximum is positive definite. Apply Section 7.1 to conclude that $\sqrt{w_\varepsilon}$ is not convex in any neighborhood of the maximum. Equivalently, $-\sqrt{w_\varepsilon}$ is not concave.

The expansion centers at the actual maximum, rather than keeping the old center fixed; this is essential for the obstruction to apply.

7.6. Ordinary strict log-concavity persists

This additional statement keeps the two concavity properties distinct. On a fixed interior compact set, (D2) and C^2 convergence preserve positive definiteness of $-D^2 \log u_\varepsilon$. Near the boundary, uniform convexity, uniform Hopf bounds and uniform $C^{2,\gamma}$ bounds supply a collar on which the same is true, uniformly for small parameter. Here is the local estimate.

In normal coordinates $x = y - d\nu(y)$, write $u_\varepsilon = a(y)d + O(d^2)$, where a is uniformly positive. At the boundary, for a unit tangent τ , $D^2u_\varepsilon[\tau, \tau] = -a \text{II}[\tau, \tau]$. The $C^{2,\gamma}$ bounds and the positive lower curvature bound therefore give, in the tangent/normal splitting at y ,

$$(-D^2 \log u_\varepsilon)_{TT} \succeq \frac{k}{d} I, \quad (-D^2 \log u_\varepsilon)_{NN} \geq \frac{k}{d^2}, \quad |(-D^2 \log u_\varepsilon)_{TN}| \leq \frac{K}{d}, \quad (\text{D18})$$

for fixed positive constants and a sufficiently small common collar. For the first estimate use that the tangential gradient term in $-D^2 \log u = -D^2 u/u + Du \otimes Du/u^2$ is nonnegative. For the mixed estimate use $|D_T u| = O(d)$ and bounded $D^2 u$; for the normal estimate use the Hopf bound and $u = ad + O(d^2)$. The Schur complement is bounded below by $k/d - K^2/k$, hence is positive for smaller d . Combining collar and interior proves strict log-concavity throughout Ω_ε .

8. Ground-state uniqueness on every bounded convex domain

This section leaves the radial setting. Throughout, $\Omega \subset \mathbb{R}^N$ is nonempty, bounded, open and convex. By [GMS, Theorem 1.1], it admits a positive log-concave solution $U \in H_0^1(\Omega) \cap C(\overline{\Omega})$ of

$$-\Delta U = U \log U^2, \quad U|_{\partial\Omega} = 0.$$

Interior elliptic regularity makes U smooth wherever it is positive. We first strengthen its concavity, then compare its energy with every function in $H_0^1(\Omega)$. In this section $w = -\log U$ is unnormalized; this changes no Hessian.

8.1. A Gaussian-strength Hessian bound without boundary regularity

Lemma 8.1. Every positive log-concave solution U as above satisfies

$$D^2(-\log U)(x) \succ I_N \quad (x \in \Omega). \quad (8.1)$$

The strict inequality is pointwise. A uniform positive gap on all of Ω is not part of the assertion.

Proof. Put $H = D^2 w$ and $\mathcal{L} = \Delta - 2\nabla w \cdot \nabla$. Direct differentiation gives

$$\Delta w = |\nabla w|^2 - 2w, \quad \mathcal{L}H = 2H^2 - 2H. \quad (8.2)$$

Initially $H \succeq 0$. For a fixed unit vector e , the nonnegative function $h = e^T H e$ satisfies

$$\mathcal{L}h = 2|He|^2 - 2h \leq (2 \operatorname{tr} H - 2)h.$$

On compact subsets the coefficients are bounded. The strong minimum principle for $\mathcal{L}h - Ch \leq 0$, with a sufficiently large nonnegative local constant C , and propagation through overlapping balls show that a zero of h forces $h \equiv 0$. Along a maximal segment parallel to e , the function w would then be affine. The segment has finite boundary endpoints, while $w \rightarrow +\infty$ there because U is continuous and zero on $\partial\Omega$. This is impossible. Hence $H \succ 0$ throughout Ω .

For $0 < t < \min\{1/2, \max U\}$ set

$$\Omega_t = \{U > t\}, \quad v = U - t, \quad w_t = -\log v.$$

The strict convexity of w gives a unique critical point, its minimum. Thus $\partial\Omega_t$ is a smooth regular level surface. Its second fundamental form is $D^2 w / |\nabla w|$ on tangent vectors, so Ω_t is uniformly convex, and $\overline{\Omega}_t \Subset \Omega$. Moreover,

$$H_t := D^2 w_t = \frac{U}{v} H + \frac{tU}{v^2} \nabla w \otimes \nabla w \succ 0. \quad (8.3)$$

On the boundary of this fixed level domain, v has a nonzero normal derivative. In an inward distance collar write $v = da$ with a smooth and positive. Then

$$D^2(-\log v) = \frac{\nabla d \otimes \nabla d}{d^2} - \frac{D^2 d}{d} - D^2 \log a.$$

Uniform convexity bounds the tangential block below by k/d and the normal block below by k/d^2 for small d , while the mixed block stays bounded. Thus $\lambda_{\min}(H_t) \rightarrow +\infty$ at $\partial\Omega_t$. In dimension one the normal estimate suffices. Its global minimum $\lambda_t > 0$ is consequently attained at an interior point x_t .

The equation for w_t is

$$\Delta w_t = |\nabla w_t|^2 + B_t(w_t), \quad B_t(s) = 2(1 + te^s) \log(e^{-s} + t). \quad (8.4)$$

Writing $u = e^{-s} + t$ and $v = u - t$, we calculate

$$B'_t(s) = -2 + \frac{2t}{v} \log u, \quad B''_t(s) = \frac{2t}{v} \left(\log u - \frac{v}{u} \right). \quad (8.5)$$

Choose a unit eigenvector e for λ_t at x_t and keep it fixed when differentiating. The scalar $e^T H_t e$ has a minimum at x_t . Differentiating (8.4) twice and using $H_t e = \lambda_t e$ gives there

$$0 \leq 2\lambda_t^2 + B'_t \lambda_t + B''_t (\partial_e w_t)^2. \quad (8.6)$$

If $U(x_t) \leq 1$, (8.5) gives $B'_t \leq -2$ and $B''_t \leq 0$. Since $\lambda_t > 0$, (8.6) implies $\lambda_t \geq 1$.

Otherwise x_t lies in the fixed compact set $K = \{U \geq 1\} \Subset \Omega$. If K is empty, the preceding case suffices. If it is nonempty, set $\delta = \min_K \lambda_{\min}(H) > 0$. Equation (8.3) gives $\lambda_t \geq \delta$ at x_t . On K we have $v \geq 1/2$ and uniform bounds on U and ∇U , so

$$|B'_t + 2| \leq Ct, \quad |B''_t| \leq Ct, \quad |\nabla w_t| \leq C.$$

When $\lambda_t < 1$, (8.6) therefore gives $2\lambda_t(1 - \lambda_t) \leq C't$, and hence $\lambda_t \geq 1 - C't/(2\delta)$. The constants are independent of t . Letting $t \downarrow 0$ at each fixed interior point proves $H \succeq I_N$.

To make the inequality strict, set $S = H - I_N \succeq 0$. From (8.2),

$$\mathcal{L}S = 2S^2 + 2S.$$

For $h = e^T S e \geq 0$ this implies $\mathcal{L}h \leq 2(\text{tr } S + 1)h$. The same strong minimum principle shows that a zero forces $h \equiv 0$. Then w has second derivative exactly one along every segment parallel to e . It is quadratic along a maximal such segment and cannot tend to infinity at its finite endpoints. This contradiction proves (8.1). $\square$

8.2. A local strict remainder in the logarithmic Sobolev inequality

For a probability measure ν , write

$$\text{Ent}_\nu(h^2) = \int h^2 \log h^2 d\nu - \left(\int h^2 d\nu \right) \log \left(\int h^2 d\nu \right).$$

Lemma 8.2. Put $Z = \int_\Omega U^2$ and $d\mu = U^2 dx / Z$. There are compact sets $K_j \Subset \Omega$, whose union has full Lebesgue measure in Ω , and constants $\delta_j > 0$ such that, for every $h \in C_c^\infty(\Omega)$,

$$\int |\nabla h|^2 d\mu - \text{Ent}_\mu(h^2) \geq \delta_j \int_{K_j} |\nabla h|^2 d\mu. \quad (8.7)$$

Proof. Let $V = -2 \log U + \log Z$. Lemma 8.1 gives $D^2 V \succ 2I_N$. Thus $W(y) = V(y) - |y|^2$, extended to $+\infty$ outside Ω , is proper, lower semicontinuous and convex. Caffarelli's contraction theorem [Caf; DF, Theorem 1.1], with the Gaussian normalization

$$d\gamma(x) = \pi^{-N/2} e^{-|x|^2} dx,$$

gives a Brenier transport $T = \nabla \Phi$ from γ to μ with

$$0 \preceq D^2 \Phi \preceq I_N \quad \text{almost everywhere.} \quad (8.8)$$

Here the theorem applies to an extended-valued convex perturbation. To see the extension explicitly, approximate W by smooth convex functions using Moreau envelopes and mollification. Retain a common affine lower bound; the resulting densities have a common integrable Gaussian upper bound and converge almost everywhere to e^{-V} , including outside $\bar{\Omega}$. Normalize the densities. Their first moments are bounded. For the corresponding contraction maps T_k ,

$$|T_k(0)| \leq \int |T_k(x)| d\gamma(x) + \int |x| d\gamma(x).$$

They therefore have a locally uniformly convergent subsequence. Its limit is a gradient of a convex function, transports γ to μ and remains a contraction. Uniqueness of the Brenier map identifies the limit as T . This argument uses no regularity of $\partial\Omega$. The later formulation [DF] explicitly credits both the original theorem and its erratum [CafE].

We justify the interior differentiations used next. Set $f = d\gamma/dx$ and $g = d\mu/dx$. The density g is globally bounded, say by M . The almost-everywhere inverse property of Brenier maps and the area formula for the Lipschitz map T give

$$\det DT(x) = \frac{f(x)}{g(T(x))} \geq \frac{\min_B f}{M} > 0 \quad \text{for almost every } x \in B \quad (8.9)$$

on any source ball B . All eigenvalues in (8.8) are at most one, so (8.9) bounds every eigenvalue below by the same positive constant on B . Consequently Φ is locally uniformly convex. Integrating its almost-everywhere Hessian bound, or first mollifying it, shows that T is strictly monotone and injective, and has a locally Lipschitz inverse. Invariance of domain makes its image open. Continuity and the transport property put its image in $\bar{\Omega}$, so in fact it lies in Ω . Its complement there has μ -measure zero.

The image of each compact source ball is compactly inside Ω . On that ball, the equation

$$\log \det D^2 \Phi = -|x|^2 + V(\nabla \Phi) + C \quad (8.10)$$

is uniformly elliptic, and its right side is locally Lipschitz. The almost-everywhere equation for the $C^{1,1}$ convex potential also holds in the viscosity sense. Interior Evans--Krylov regularity and Schauder

bootstrapping give $\Phi \in C^\infty$; see [CC] for the interior concave-equation regularity used here. For applying uniform ellipticity, one can extend log det from the compact interval of positive Hessians supplied by (8.8)–(8.9) to a uniformly elliptic concave operator. No boundary regularity or pointwise surjectivity of T is required.

In fact $D^2\Phi \prec I_N$ everywhere. If $\Phi_{ee}(x_0) = 1$ for a unit eigenvector e , the function $\Phi_{ee} \leq 1$ has a maximum at x_0 . Put $A = D^2\Phi$ and differentiate (8.10) twice in direction e :

$$\begin{aligned} \operatorname{tr}(A^{-1}D^2\Phi_{ee}) - \operatorname{tr}(A^{-1}(\partial_e A)A^{-1}(\partial_e A)) &= -2 + D^2V(T)[Ae, Ae] \\ &\quad + \nabla V(T) \cdot \nabla \Phi_{ee}. \end{aligned} \quad (8.11)$$

At x_0 the left side is nonpositive. On the right, $Ae = e$, $\nabla \Phi_{ee} = 0$, and $D^2V \succ 2I_N$, making it strictly positive. This contradiction proves the strict contraction. Let $K_j = T(\overline{B_j})$. These compact sets cover the image of T , a full-measure subset of Ω , and for some $\delta_j > 0$,

$$I_N - DT(x)^2 \succeq \delta_j I_N \quad (x \in \overline{B_j}). \quad (8.12)$$

The Gaussian logarithmic Sobolev inequality [Gross], in the chosen normalization, is

$$\operatorname{Ent}_\gamma(q^2) \leq \int |\nabla q|^2 d\gamma. \quad (8.13)$$

Its constant can be checked by the Ornstein–Uhlenbeck semigroup P_s with generator $\Delta - 2x \cdot \nabla$. For smooth bounded positive F bounded away from zero, invariance, integration by parts and the gradient commutation identity give

$$\begin{aligned} \operatorname{Ent}_\gamma(F) &= \int_0^\infty \int \frac{|\nabla P_s F|^2}{P_s F} d\gamma ds, \\ \frac{|\nabla P_s F|^2}{P_s F} &\leq e^{-4s} P_s \left(\frac{|\nabla F|^2}{F} \right). \end{aligned}$$

Integration yields $\operatorname{Ent}_\gamma(F) \leq \frac{1}{4} \int |\nabla F|^2 / F d\gamma$. Applying this to $q^2 + \varepsilon$ and using approximation proves (8.13), including bounded Lipschitz q . These are classical facts, recalled to fix the coefficient.

Apply (8.13) to $q = h \circ T$ and use change of variables. By the chain rule and (8.12),

$$\begin{aligned} \int |\nabla h|^2 d\mu - \operatorname{Ent}_\mu(h^2) &\geq \int \nabla h(T)^T (I_N - DT^2) \nabla h(T) d\gamma \\ &\geq \delta_j \int_{K_j} |\nabla h|^2 d\mu. \end{aligned}$$

The sets K_j have full-measure union because $g > 0$ in Ω . This proves (8.7). $\square$

8.3. A sharp energy inequality and its equality case

For real $v \in H_0^1(\Omega)$ define

$$\mathcal{F}(v) = \int_\Omega |\nabla v|^2 - \int_\Omega v^2 \log v^2, \quad m(v) = \int_\Omega v^2, \quad (8.14)$$

with $0 \log 0 = 0$. Both integrals are finite on a bounded domain.

Proposition 8.3. For every nonzero real $v \in H_0^1(\Omega)$,

$$\mathcal{F}(v) + m(v) \log \frac{m(v)}{Z} \geq 0, \quad (8.15)$$

with equality if and only if $v = cU$ for some nonzero real constant c .

Proof. First let $\phi \in C_c^\infty(\Omega)$ and $h = \phi/U$. Testing the weak equation of U with ϕ^2/U gives the ground-state identity

$$\int_{\Omega} U^2 |\nabla(\phi/U)|^2 = \int_{\Omega} |\nabla \phi|^2 - \int_{\Omega} \phi^2 \log U^2. \quad (8.16)$$

Writing $m = \int \phi^2$, the definition of entropy then gives exactly

$$Z \left(\int |\nabla h|^2 d\mu - \text{Ent}_{\mu}(h^2) \right) = \mathcal{F}(\phi) + m \log(m/Z). \quad (8.17)$$

Thus Lemma 8.2 proves, for every j ,

$$\mathcal{F}(\phi) + m \log(m/Z) \geq \delta_j \int_{K_j} U^2 |\nabla(\phi/U)|^2. \quad (8.18)$$

Now take $\phi_n \rightarrow v$ strongly in $H_0^1(\Omega)$. Choose a small $\eta > 0$ with $2 + \eta$ strictly below the Sobolev exponent. The bound

$$|s^2 \log s^2| \leq C_{\eta}(|s|^{2-\eta} + |s|^{2+\eta})$$

and strong Sobolev convergence show uniform integrability and hence convergence of the entropy integrals. The same argument uses any suitable finite Sobolev exponent when $N \leq 2$. Consequently the left side of (8.18) converges to $\mathcal{F}(v) + m(v) \log(m(v)/Z)$. On a neighborhood of each K_j , division by the smooth positive U is continuous in H^1 , so the right side converges as well. Formula (8.18) holds for every $v \in H_0^1(\Omega)$.

This passage does not assume that v/U has finite global weighted Dirichlet energy or that $v^2 \log U$ is integrable. Only the compactly supported identity (8.16) and convergence on compact subsets are used. In particular (8.15) holds. If its left side vanishes, (8.18) forces $\nabla(v/U) = 0$ on every K_j , hence almost everywhere in Ω . Since $v/U \in H_{\text{loc}}^1(\Omega)$ and Ω is connected, $v = cU$.

Conversely testing the equation with U gives $\mathcal{F}(U) = 0$. Direct scaling yields $\mathcal{F}(cU) = -c^2 Z \log c^2$, so every nonzero multiple attains equality in (8.15). $\square$

8.4. Ground states and the remaining uniqueness question

The action corresponding to the equation is

$$\mathcal{J}(v) = \frac{1}{2} \mathcal{F}(v) + \frac{1}{2} m(v). \quad (8.19)$$

It is C^1 on $H_0^1(\Omega)$; its critical points are the weak solutions. The nonzero Nehari set is $\mathcal{N} = \{v \neq 0 : \mathcal{F}(v) = 0\}$. On this set $\mathcal{J}(v) = m(v)/2$. This agrees with the ground-state definition in [GMS, Section 4.3], restricted there to nonnegative functions.

By (8.15), every $v \in \mathcal{N}$ has $m(v) \geq Z$. The existing solution U attains Z and therefore minimizes the action on $\mathcal{N}$. If equality holds, Proposition 8.3 gives $v = cU$, and $\mathcal{F}(v) = 0$ forces $c = \pm 1$. Thus the real Nehari minimizers, and the real ground states, are precisely U and $-U$. In particular the positive ground state is unique. This proves Theorem 1.5.

The same proposition gives the equivalent constrained characterization

$$\min_{\substack{v \in H_0^1(\Omega) \\ \|v\|_2=1}} \mathcal{F}(v) = \log Z, \quad \text{Argmin} = \{U/\sqrt{Z}, -U/\sqrt{Z}\}. \quad (8.20)$$

Every positive log-concave solution satisfies Lemma 8.1 and the ensuing comparison, so every such solution is this unique ground state. In particular the solutions in Theorem 1.4 are ground states on their respective deformed domains. Their loss of normalized square-root-log concavity therefore occurs already among ground states.

This answers the ground-state uniqueness and ground-state identification questions in [GMS, Remark 1.6]. The existence theorem of [GMS] remains an essential input. Uniqueness among all positive solutions would additionally require excluding positive solutions of greater action; nothing above supplies that exclusion.

9. Further questions

Uniqueness among all positive solutions on an arbitrary bounded convex domain remains open here: Theorem 1.5 rules out additional ground states, but does not exclude positive solutions of higher energy. The optimal positive-power exponent on balls in higher dimensions remains implicit, and its radius dependence is unresolved. Theorem 1.4 settles the normalized square-root-log deformation question negatively, now for ground states by Theorem 1.5; the weaker property allowing sufficiently small rescaling remains a separate question.

References

- [BC] M. Ben Chrouda, *Uniqueness and Liouville type results for radial solutions of some classes of k -Hessian equations*, Electronic Journal of Qualitative Theory of Differential Equations **2022**, no. 62, 1--17. [doi:10.14232/ejqtde.2022.1.62](https://doi.org/10.14232/ejqtde.2022.1.62).
- [GMS] M. Gallo, S. Mosconi and M. Squassina, *Power law convergence and concavity for the logarithmic Schrödinger equation*, Mathematische Annalen **395** (2026), article 21, 60 pp. [doi:10.1007/s00208-026-03452-2](https://doi.org/10.1007/s00208-026-03452-2). [arXiv:2411.01614v2](https://arxiv.org/abs/2411.01614v2).
- [GMoS] M. Gallo, R. Moraschi and M. Squassina, *Quantitative and exact concavity principles for parabolic and elliptic equations*, [arXiv:2504.09494v2](https://arxiv.org/abs/2504.09494v2). Also compared with the [41-page author manuscript](#), accessed September 15, 2026.
- [LSZ] T. Liu, X. Sun and W. Zou, *Uniqueness of bound states to the logarithmic Schrödinger equation*, [arXiv:2606.19077v2](https://arxiv.org/abs/2606.19077v2), June 22, 2026.
- [IST] K. Ishige, P. Salani and A. Takatsu, *New characterizations of log-concavity*, [arXiv:2004.13381](https://arxiv.org/abs/2004.13381), Section 4.2. The distinction between fixed normalization and concavity after small rescaling is used here; no heat-flow theorem is a proof dependency.
- [KS] D. J. Knezevic and E. Suli, *Spectral Galerkin approximation of Fokker-Planck equations with unbounded drift*, ESAIM: Mathematical Modelling and Numerical Analysis **43** (2009), 445--485.

[Primary journal PDF](#), Theorem 2.1. The classical Brascamp-Lieb inequality is credited; the needed ball boundary version is also proved above.

[Caf] L. A. Caffarelli, *Monotonicity properties of optimal transportation and the FKG and related inequalities*, Communications in Mathematical Physics 214 (2000), 547--563. DOI: 10.1007/s002200000257.

[CafE] L. A. Caffarelli, *Erratum: Commun. Math. Phys. 214, 547--563 (2000)*, Communications in Mathematical Physics 225 (2002), 449--450. DOI: 10.1007/s002200100560.

[DF] G. De Philippis and A. Figalli, *Rigidity and stability of Caffarelli's log-concave perturbation theorem*, Nonlinear Analysis 154 (2017), 59--70. DOI: 10.1016/j.na.2016.10.006. Theorem 1.1 restates the contraction theorem, citing both [Caf] and [CafE].

[Gross] L. Gross, *Logarithmic Sobolev inequalities*, American Journal of Mathematics 97 (1975), 1061--1083.

[CC] L. A. Caffarelli and X. Cabré, *Fully Nonlinear Elliptic Equations*, American Mathematical Society Colloquium Publications 43, 1995.

Preparation note

Henry Zweiman. Revision 1.2, September 16, 2026. This revision adds Theorem 1.5 and Section 8, proving ground-state uniqueness on every bounded convex domain and identifying every log-concave solution as that ground state. The ball and deformation proof bodies in Sections 2--7 are retained. The argument is by hand; no finite computation is used as a proof of a universal claim. Proof and source checks are internal and do not constitute independent expert review, exhaustive priority certification or journal acceptance.